

1 Marginal Probability Mass Function (PMF)

For discrete random variables X and Y , the joint behavior is described by the joint PMF $p_{XY}(x, y) = P(X = x, Y = y)$ [cite: 73, 74].

1.1 Definition

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable [cite: 77, 78]:

- $p_X(x) = \sum_y p_{XY}(x, y)$ [cite: 75, 76]
- $p_Y(y) = \sum_x p_{XY}(x, y)$ [cite: 76]

1.2 Marginal PMF from a Joint PMF Table

In a tabular representation, the row sums provide $p_X(x)$, and the column sums provide $p_Y(y)$ [cite: 85, 93].

- $p_X(x_i) = \sum_j p_{ij}$ and $\sum_i p_X(x_i) = 1$ [cite: 96]
- $p_Y(y_j) = \sum_i p_{ij}$ and $\sum_j p_Y(y_j) = 1$ [cite: 97, 98]

2 Correlation

Correlation is a basic way to measure the joint behavior of two random variables through the "product moment" [cite: 133, 134].

2.1 Definition

In the continuous case, the raw correlation R_{XY} is defined as [cite: 135, 136]:

$$R_{XY} = E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y) dx dy$$

[cite: 137]

- It represents the average value of the product XY [cite: 138].
- If $R_{XY} = 0$, the variables are considered orthogonal about the origin [cite: 150, 151].
- **Limitation:** It depends on the scale of X and Y and mixes mean values, spread, and dependence [cite: 144, 146, 148, 149].

3 Covariance

To create a cleaner measure of dependence, we center the variables by subtracting their means [cite: 152, 154].

3.1 Definition and Interpretation

Covariance measures how centered quantities vary together[cite: 165]:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

[cite: 162] where $\mu_X = E[X]$ and $\mu_Y = E[Y]$ [cite: 163].

- **Positive Covariance** (> 0): Deviations tend to have the same sign (e.g., as study hours increase, GPA increases)[cite: 172, 175, 193].
- **Negative Covariance** (< 0): One variable increases while the other decreases (e.g., as TV time increases, GPA decreases)[cite: 173, 176, 194].
- **Zero Covariance** ($= 0$): No linear co-variation trend[cite: 174, 177].

3.2 Algebraic Form and Properties

The computational formula for covariance is[cite: 216, 227]:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

[cite: 229]

Key Properties:

1. **Symmetry:** $\text{Cov}(X, Y) = \text{Cov}(Y, X)$ [cite: 235].
2. **Variance as Special Case:** $\text{Cov}(X, X) = \text{Var}(X)$ [cite: 244, 253].
3. **Linearity:** $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$ [cite: 255, 265].
4. **Independence:** $X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$ (Independence implies zero covariance, but the reverse is not always true)[cite: 267, 268, 269].

4 Pearson's Correlation Coefficient (ρ_{XY})

The Pearson coefficient is the normalized version of covariance[cite: 353, 386].

4.1 Definition

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

[cite: 355, 384, 392]

- **Range:** Always between -1 and 1 ($-1 \leq \rho_{XY} \leq 1$)[cite: 354, 397].
- **Unit-free:** It is independent of the units of X and Y [cite: 356, 387].
- **Strength:** Unlike covariance, it measures both the direction and the *strength* of the linear relationship[cite: 350, 359, 393].

5 Jointly Gaussian Random Variables

A pair (X, Y) is jointly Gaussian if its joint density has a bivariate Gaussian form and the marginals are Gaussian[cite: 568, 576].

5.1 Joint PDF Form

For two variables X and Y [cite: 589]:

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) \right]\right)$$

[cite: 590, 606, 634]

5.2 Applications

- **Signal Processing & Communication:** Natural model for continuous noisy measurements [cite: 580, 581, 582].
- **Climate Modeling:** Used to study relationships like CO_2 concentration and temperature anomalies [cite: 604, 605, 615].

Summary for Exam Revision

Concept	Key Formula / takeaway
Marginal PMF	Sum the joint PMF over the "other" variable. [cite: 78]
Correlation (R_{XY})	$E[XY]$. Measures raw joint behavior. [cite: 136]
Covariance (Cov)	$E[XY] - E[X]E[Y]$. Measures linear direction. [cite: 227]
Pearson's (ρ)	$\frac{Cov(X,Y)}{\sigma_X\sigma_Y}$. Measures direction AND strength (-1 to 1). [cite: 355]
Independence	If independent, $Cov = 0$ and $\rho = 0$. [cite: 268, 558]
Joint Gaussian	Marginals are Gaussian; defined by μ, σ, ρ . [cite: 568, 649]